Materials and Design 30 (2009) 3308–3312

Contents lists available at ScienceDirect

Materials and Design

journal homepage: www.elsevier.com/locate/matdes

Short Communication

Investigation of process parameters effect on the copper-wire drawing

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ARTICLE INFO

Article history:
Received 25 September 2008
Accepted 5 December 2008
Available online 13 December 2008

ABSTRACT

In the present paper, the copper-wire drawing process is analysed in order to optimise the die design and to obtain the best quality of wire. Different design parameters of the drawing process have been studied using FE-analysis and experiment. This study investigates the effect of the process variables such as semi-die angle and reduction in area, and the coefficient of friction on the drawing force value. For the wire, elastoviscoplastic behaviour is considered, while the die material is assumed to be elastic. Two types of dies have been used in this work: crystal mono diamond core (MD) and crystal poly diamond (PCD).

Experiments were carried out on drawing machine involving industrial conditions to measure the drawing force. The obtained results have been compared with analytical methods available in literature, particularly, with Siebel, Wistreich and Avitzur's solutions.

Our results clearly indicate that friction has significant effects on the drawing force which decrease with the decrease of area reduction. The optimum die angle for wire drawing is assumed to be obtained when the plastic strain distribution across the diameter of the wire becomes uniform.

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1. Introduction

Drawing is a metalworking process of bars, rods and wires particularly used in the electric and automotive sectors. The process consists of reducing the cross-section by forcing the wire through series of dies. During the drawing operation, a plastic deformation is imparted to the material depending on the drawability of the material. Information about the mechanical deformation is required in order to design dies and processing parameters. The main process parameters are: the die semiangle "alpha" to be defined in Figs. 6 and 7, the cross-sectional area reduction, r, and the friction coefficient, μ, this latter represents the friction along the die-wire interface. These parameters have a major influence over the final wire quality, the drawing force, the lubrication in the process, the mechanical properties and the wear of the dies. Results of research works already published indicate different approaches to solve of such problems.

Theoretical solutions presented in the literature principally concern the drawing process based on the relationship between the drawing stress and process parameters. The deformation during wiredrawing results from three components: a homogeneous deformation depending on the reduction ratio, an inhomogeneous deformation which depends on the geometrical parameters and friction in the die [1]:

$$\frac{\sigma_f}{\sigma_0} = \ln\left(\frac{R_i}{R_f}\right) + \mu\cot\alpha\ln\left(\frac{R_i}{R_f}\right) + \frac{2\alpha}{3}, \quad (1)$$

where σ_f is the applied stress or drawing stress, σ_0 is the yield stress, R_i is the wire radius ahead of the die, R_f is the wire radius behind the die, α the die semi-angle and μ the friction coefficient. Eq. (1) can be written

$$\frac{\sigma_f}{\sigma_0} = A + \frac{2\alpha}{3}, \quad (2)$$

where

$$A = \ln\left(\frac{R_i}{R_f}\right) + \mu\cot\alpha\ln\left(\frac{R_i}{R_f}\right).$$

From this decomposition, an analytical formulation has been proposed to describe the evolution of the drawing stress σ_f according to geometrical parameters, the friction coefficient and the plastic deformation [2–4]:

$$\frac{\sigma_f}{\sigma_0} = \Phi\left(\ln\left(\frac{R_i}{R_f}\right) + \mu\cot\alpha\ln\left(\frac{R_i}{R_f}\right)\right), \quad (3)$$

where the quantity Φ depends on reduction, semiangle and hardening parameters, but not on coefficient of friction. Eq. (3) can be expressed as:

$$\frac{\sigma_f}{\sigma_0} = A\Phi. \quad (4)$$

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doi:10.1016/j.matdes.2008.12.006

Note that the drawing stress can be obtained by adding the two terms in Siebel's equation (Eq. (2)) or by multiplying them in Wistreich's equation (Eq. (4)).

Using the assumption of Coulomb's coefficient of friction between the die and wire, Avitzur [5–7] has proposed the relative drawing stress which is built up from these four components: the power due to drawing force (W_f), the power required to overcome the back tension (W_b), the internal power of deformation (W_i) and the power needed to overcome friction losses (W_s).

By writing the work balance in the following way: $W_s + W_b + W_i - W_f = 0$, the author proposes a relation between the ratio of drawing stress σ_f to yield stress σ_0 and various variables of the process:

$$\frac{\sigma_f}{\sigma_0} = \frac{\left[\frac{\sigma_b}{\sigma_0} + 2f(\alpha)\ln\left(\frac{R_i}{R_f}\right) + \frac{2}{\sqrt{3}}\left(\frac{\alpha}{\sin^2\alpha} - \cot\alpha\right) + 2\mu\left(\cot(\alpha)\left(1 - \frac{\sigma_b}{\sigma_0} - \ln\left(\frac{R_i}{R_f}\right)\right)\ln\left(\frac{R_i}{R_f}\right) + \frac{P}{R_f}\right)\right]}{\left[1 + 2\mu\frac{P}{R_f}\right]}, \quad (5)$$

where σ_b is the back tension and P is the bearing length.

Using the finite element method, several researchers have investigated optimal wire drawing conditions in relation with various factors (such as die angle, reduction, friction coefficient, or bearing length). The accuracy of finite element simulation of wiredrawing is generally related to material characteristics and geometrical parameters. Chen et al. [8] obtained the steady state deformation characteristics in extrusion and wiredrawing as functions of material properties, the die work interface friction and the die angle and reduction. They concluded that the finite element results obtained in extrusion could not be extrapolated to obtain results in drawing. Chin and Steif [9] used the numerical method to study wire stress-strain behaviour and relations between drawing force and straining. The effects of various process variables (the reduction ratio, the die semi-angle, the coefficient of friction and the back tension) on important drawing parameters (die pressure, the drawing stress) have been investigated by Dixit and Dixit [10]. They observed that an increase in the reduction, or a decrease in the die semi-angle, or coefficient of friction tends to make the product more homogeneous. Kim et al. [11] analysed the wiredrawing process using rigid-plastic finite element method to determine the main parameters affecting the die wear.

Recently, several studies about metalforming processes have used the finite element simulation in order to calculate the drawing force and obtain strains, stresses, or contact pressure distributions. Eivani and Karimi Taheri [12] have studied the effect of die angle, friction coefficient and the dead metal zone on plastic deformation using upper bound method. A comparative study between analytical methods and FEM results is presented by Luis et al. [13] in order to obtain forces, energies, stresses and strains which are involved in the wire drawing process. Chen and Huang [14] have employed the finite element method and the Taguchi methods to optimise the process parameters of the wire drawing process. The results have shown that the semi-angle of the die, the friction factor, the length of the bearing part of the die, and the temperature have a significant effects on the drawing process. Two materials, copper and steel, have been tested under different friction conditions, die angles and drawing velocities by Majzoobi et al. [15], who observed that the optimum die angle slightly decreases with the increase of drawing velocity. Heng-Sheng et al. [16] have investigated the effect of Δ-parameter ($\Delta = h/L$ where h is the mean diameter of the workpiece and L is the contact length between the deforming workpiece and the die) on inhomogeneous deformation and residual stress of mid-carbon steel through various combinations of area reduction and die semi-angle.

In the present work, both experimental and numerical approaches are carried out to obtain the optimum conditions for the minimum drawing force and, consequently, reducing the cost of the process. The experiments are conducted using the multipass drawing machine involving industrial conditions and the obtained results are compared with analytical methods available in the literature. Note that, numerical simulations are carried out using ANSYS code [17]. The effects of different process parameters on the drawing force and the equivalent strain distribution across the diameter of the wire, such as the die angle, the reduction areas and the friction between the wire and the drawing die, are investigated.

2. Experimental procedure

2.1. Tensile test

A uniaxial test is carried out in various speeds (0.5, 10 and 20 mm/min) in order to study the copper wire (diameter 5.13 mm) elasto-viscoplastic behaviour. The stress–strain curves (Fig. 1) show that the variation of the strain rate changes significantly the material behaviour in the tensile test. With the range of speed, the strain rate varies between 0.083×10^{-3} and $3.33 \times 10^{-3}\ s^{-1}$. The initial strain rate for speed of 0.5 mm/min is small enough to assume that the material behaviour is similar to that of a quasi-static testing. The isotropic hardening behaviour is implemented as:

$$\sigma = \sigma_{y0} + K(\varepsilon_p)^n, \quad (6)$$

where σ_{y0} is the static yield stress ($\sigma_0 = 166$ MPa). The hardening parameters are the strength coefficient K ($K = 484$ MPa) and the hardening exponent n ($n = 0.82$). Actually, the plastic behaviour of the copper wire is simulated by FEM, and values of plastic strains are results. To model the viscoplastic behaviour of the wire, the Perzyna rate-dependent model is used [18]:

$$\sigma_y = \left[1 + \left(\frac{\dot{\varepsilon}_p}{\gamma}\right)^m\right]\sigma_{y0}, \quad (7)$$

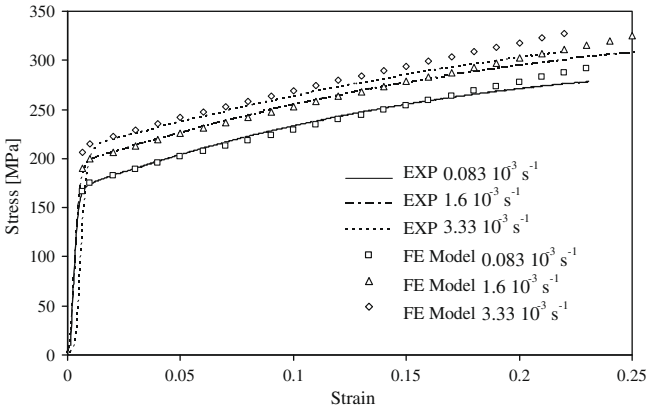


Fig. 1. Stress–strain curves of copper wire in various speeds.

where σ_y is the yield stress with respect to plastic strain rate $\dot{\varepsilon}_p$, material viscosity parameter γ, strain rate hardening parameter m and the static yield stress σ_{y0}. By fitting the Perzyna equation on the experimental data, we obtain m and γ as: $m = 0.737$ and $\gamma = 2.3$.

2.2. Drawing test

A multipass drawing machine is used for this purpose to carry out tests involving industrial conditions which imply the instrumentation of the process on the industrial machine and to measure the drawing force (Fig. 2). The reason for this is that the drawn wire must have a sufficient resistance and cross-section to withstand the drawing effort without breaking. In addition, the range of the copper wires tested corresponds to geometries usually used in the industry and the proposed test requires manufacturing of dies (PCD or diamond) which are used for particular applications, especially for difficult drawings and special schedules. For other process parameters, the industrial test is more complicated due to the damage of wire, material cost and time consuming experiments.

In this paper, the machine consists in 19 capstans, around of which the wire is wound between two consecutive drawing operations and the 21 dies which are the principal equipment used for metal forming. The friction between the wire and the capstan is created by this winding. The whole drawing system (capstans, dies) is fully immersed in the lubricant and wire rolling gauge, placed before the die, is used to measure the drawing force. Wire drawing is an interaction of several parameters, yield stress, modulus of elasticity, work hardening (wire properties) and semi-angle, reduction in area, bearing length (die geometry). An experimental test is used to carry out the wiredrawing of a copper wire through dies for various geometries. A copper material is drawn from a diameter 0.513 to 0.403 mm (reduction ratio $r = 0.375$) with a velocity of 1 m/s.

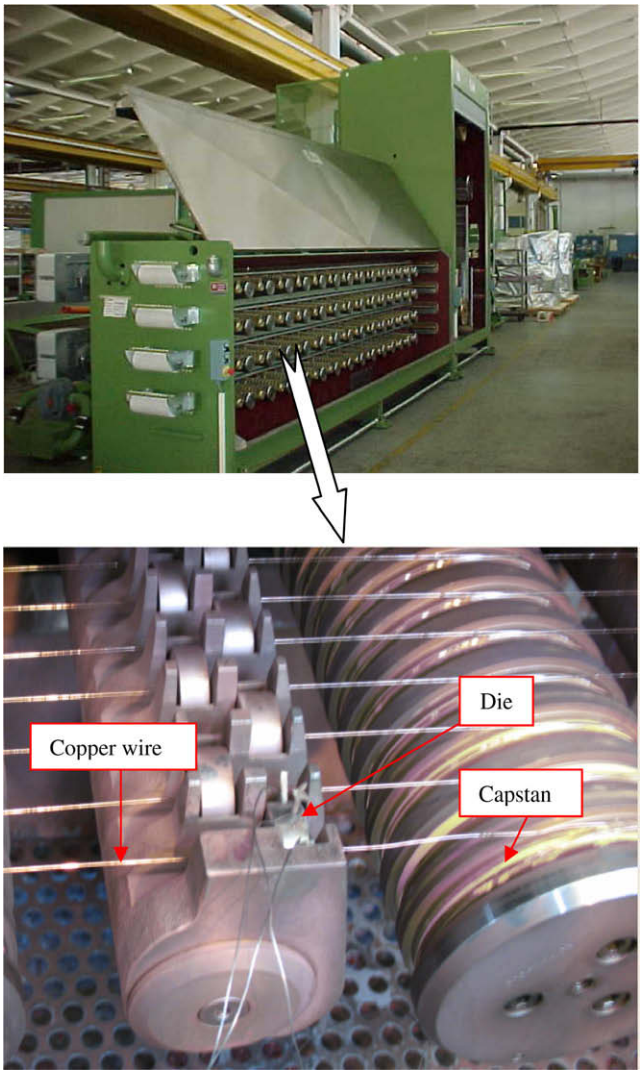


Fig. 2. Multipass drawing machine and principal equipment.

Table 1
Wiredrawing force according to the parameters.

Diameter (mm)	0.405							
Die materials	MD	MD	MD	MD	PCD	PCD	PCD	PCD
Bearing length P (% diameter)	25	25	40	40	25	25	40	40
Semiangle 2α (°)	14.90	16.85	15.60	19.05	15.05	18.45	14.75	17.25
	Drawing force (N)							
Experimental	35	37	35	34	41	37	37	41
Siebel [1]	35.91	33.95	35.42	33.72	35.80	34.7	36.02	34.48
Wistreich [2,3]	32.6	29.84	31.95	29.47	32.45	30.94	32.74	30.64
Avitzur [5–7]	34.07	33.14	34.30	33.60	34.01	33.47	34.57	33.89

During the reduction in the drawing cone, the wire affects the die by a shear stress along the drawing cone, and a pressure to the cone. Two types of dies, crystal mono diamond core (MD), or crystal poly diamond (PCD), are used in the wire drawing industry. A laser system of measurement enables us to obtain the internal geometries values of the dies. Various tests are carried out for different semiangles 2α varying from 14.75° to 19.05°) and two values of bearing (25% and 40% of the final diameter). The wire and dies are immersed in a lubricant made of oil. The wiredrawing force, measured with a deformation gauge, varies from 35 to 41 N for the various studied configurations. It seems to be a small variation of drawing force with process variables. In Table 1, the experimental results are compared with those obtained by Siebel, Wistreich, and Avitzur equations. It is observed that there is good agreement between our own experimental results and those of Siebel and Avitzur. Wistreich's model gives an underestimation compared to Siebel's and Avitzur's because the inhomogeneous deformation is not taken into account in Eq. (3). The friction coefficient μ varying between 0.1 and 0.2 corresponds to the interval of values usually observed in the case of the used lubrication type [19–21]. It should be noted that, for a given configuration (the bearing length $P = 40$ %), the wiredrawing force is slightly higher for poly crystal (PCD) dies than that of diamond (MD). For experimental results obtained here, the selected parameters, corresponding to the used industrial specifications for the wiredrawing of copper, are close to the conditions minimizing the force.

3. Numerical simulation

3.1. Finite element model

The finite element models are validated for elasto-plastic and elasto-viscoplastic material behaviours. To study the influence of some parameters such as the friction coefficient, the semiangle, the reduction ratio on the drawing stress, the wire drawing process is modelled taking into account the strain rate effect.

In the present study, at the die-wire interface, the friction is modelled by Coulomb's law, and it is chosen in the range 0.05–0.15. The wire is represented by an axisymmetric finite element model and it is moved through the die (see Fig. 3). Consequently, the following boundary conditions were chosen: symmetry conditions along the y-axis and an applied velocity at the end of the wire. The wire is modelled for each diameter with 10 elements

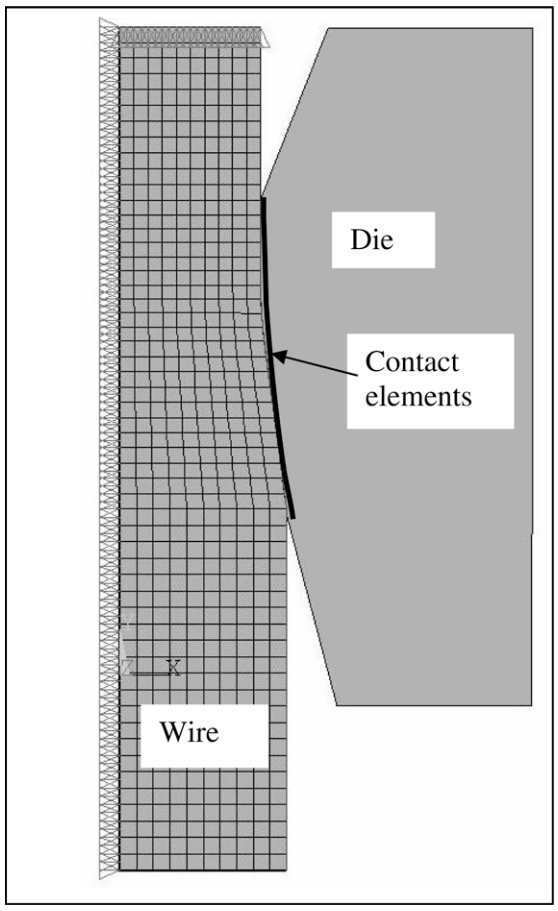


Fig. 3. Finite element model.

along the radius. The die is considered as a rigid body. Several computations have been performed to calculate the drawing force and to investigate how the drawing practice affects the plastic flow and the strain–stress distribution inside the deformation zone during the drawing of the copper wire.

3.2. Results of FE-analysis

Fig. 4 shows the result of the wiredrawing force according to the semi-angle calculated numerically in $P = 40\%$ configuration with $r = 0.375$ and for various values of μ. It can be seen that the optimum die-angle increases with increasing coefficient of friction. Further, the variation of drawing force with coefficient of friction is more significant at smaller die-angles in comparison with greater die-angles. Now, supposing that the coefficient of friction is not known accurately in a process, the die is to be designed according to maximum expected coefficient of friction and by choosing the optimum value of α according to Fig. 4.

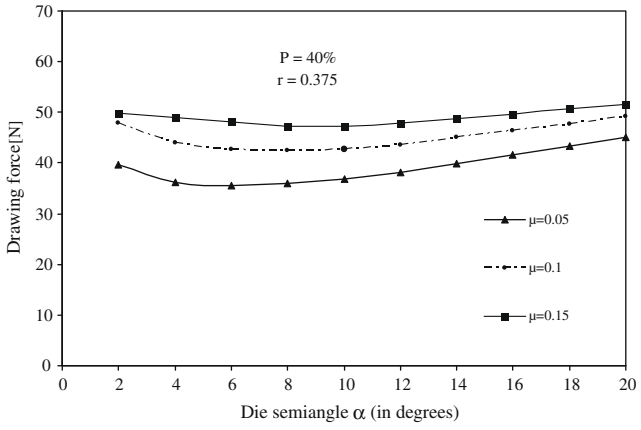


Fig. 4. Drawing force obtained from FEM analysis.

Fig. 5 depicts the variation of the wiredrawing force obtained numerically for $\mu = 0.05$ and $P = 40\%$, while varying the reduction ratio from 0.2 to 0.35. It is noted that the convexity of the curve is more significant at greater reduction ratio. At small die-angles, the drawing force is high due to the increase of the contact length between the wire and die, but at greater die-angles, the drawing force is high because of the large distortion. In Fig. 4, minimum drawing forces are seen with 5 and 8 die semi-angle degrees and friction coefficients 0.05 and 0.15, respectively, while the drawing force is minimum under 7° and friction coefficient 0.1. It is observed, in Fig. 5, that a die semi angle of 10 degrees gives the minimum drawing force at a coefficient of friction 0.05 and reductions 0.25, 0.3 and 0.35. In the problem of extrusion, a similar effect was observed by Danishevs'kyy et al. [22].

In the wire drawing process, the cross section of wire through dies is reduced by the plastic deformation. The results of equivalent plastic strain are shown in Figs. 6 and 7, where the strain is plotted as a function of radial position. The coefficient of friction has been fixed at 0.1 and three die semi-angles have been used.

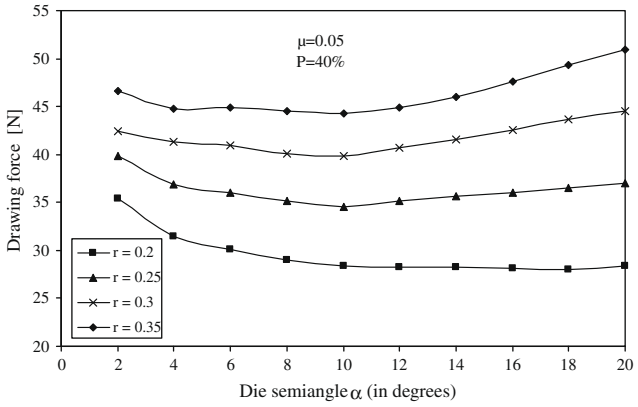


Fig. 5. Variation of drawing force with die-angle at different reductions.

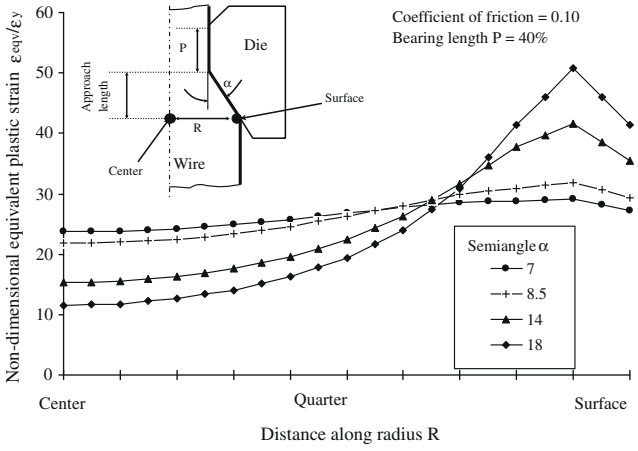


Fig. 6. Ratio of equivalent plastic strain to initial yield strain versus distance along radius at the entrance of the die.

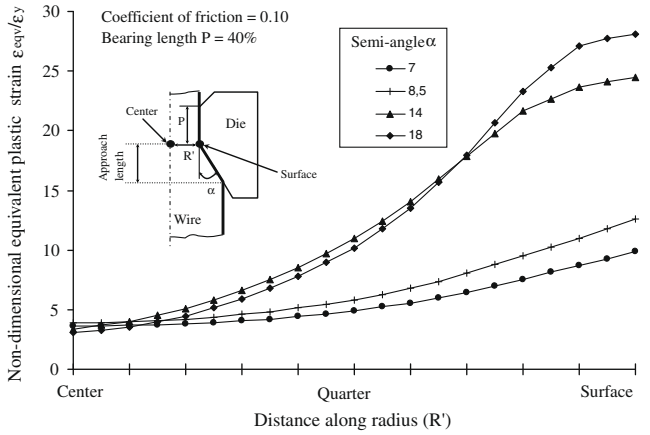


Fig. 7. Ratio of equivalent plastic strain to initial yield strain versus distance along radius at the exit of the die.

We can see that the equivalent strain has a peak value near the surface, and that this peak value diminishes as the die semiangle decreases. The obtained solutions have been contrasted with experimental results given by Wistreich [3] and with the FEM simulation proposed by Oeverstam [23] for high carbon steel. The die angle at which the drawing stress takes a minimum value is called the optimum die angle. The strain distribution is generally uniform along the wire radius for die semiangle $\alpha = 7^\circ$ and $\alpha = 8^\circ$, whereas it is not uniform at higher semiangle $\alpha = 14^\circ$ and $\alpha = 18^\circ$.

4. Conclusions

The wire drawing process of copper has been studied using experimental and numerical approaches. The material is considered to be viscoplastic and a constant coefficient of friction is used at the die–wire interface. The results have shown that, the die angle, the friction coefficient, and the bearing length, have significant effects on the drawing force during the drawing process of copper. In this paper, we have found that the drawing force increases when the reduction and the die angle increase. The coefficient of friction has an influence over the drawing force. The experiment results obtained from drawing of copper show a close correlation with values from literature. The FE simulation is applied to evaluate different types of situations in order to predict the evolution of important wire design parameters during the drawing process. It is observed that the maximum equivalent strain increases significantly with the values of the die angle. The effect of the increase of the strain makes the deformation more inhomogeneous over the cross section, principally near the die wire interface, causing the damage of wire.

Acknowledgements

The present research work has been supported by the NEXANS Metallurgy Centre, France; the authors gratefully acknowledge this institution for their support. The authors would like to thank Mr. Francis DEBLADIS, Mr. Michel SAUREL and Mr. Mickael VEAUX, for their fruitful discussions and their comments.

References

[1] Siebel E. Der derzeitige Stand der Erkennisse uber die Mechanischen Vorgange Beim Drahtziehen. Stahl und Eisen Heft ll/l2. 1947;66/67:171–80.
[2] Wistreich JG. Investigation of mechanics of wire drawing. Proc Inst Mech Eng 1955;169:654–70.
[3] Wistreich JG. The fundamentals of wire drawing. Metall Rev 1958;3–10: 97–142.
[4] Wright RN. Mechanical analysis and die design. Wire J Int 1979:60.
[5] Avitzur B. Metal forming: processes and analysis. New York: McGraw-Hill Book Co.; 1968. also, Krieger, Huntington, NY; 1979.
[6] Avitzur B. Study of flow through conical converging dies. In: Hoffmanner AL, editor. Metal forming: interrelation between theory and practice. New York: Plenum Press; 1971. p. 1–46.
[7] Avitzur B. Flow through conical converging dies. Handbook of metal forming. New York: John Wiley & Sons Inc.; 1983 [Chapter 3].
[8] Chen CC, Oh SI, Kobayashi S. Ductile fracture in axisymmetric extrusion and drawing. Part1: Deformation mechanics of extrusion and drawing, Part2: Workability in extrusion and drawing. Trans ASME J Eng Ind 1979;101:23–44.
[9] Chin RK, Steif PS. A computational study of strain inhomogeneity in wire drawing. Int J Mach Tools Manuf 1995;35:1087–98.
[10] Dixit US, Dixit PM. An analysis of the steady-state wire drawing of strain hardening materials. J Mater Process Technol 1995;47:201–29.
[11] Kim BM, Kim TH, Choi JC. Prediction of die wear in the wire-drawing process. J Mater Process Technol 1997;65(1/3):11–7.
[12] Eivani AR, Karimi Taheri A. The effect of dead metal zone formation on strain and extrusion force during equal channel angular extrusion. Comput Mater Sci 2008;42:14–20.
[13] Luis CJ, Leon J, Luri R. Comparison between finite element method and analytical methods for studying wire drawing processes. J Mater Process Technol 2005;164/165:1218–25.
[14] Chen DC, Huang JY. Design of brass alloy drawing process using Taguchi method. Mater Sci Eng 2007;464:135–40.
[15] Majzoobi GH, Fereshteh Saniee F, Aghili A. An investigation into the effect of redundant shear deformation in bar drawing. J Mater Process Technol 2008; 201:133–7.
[16] Heng-Sheng L, Yuan-Chuan H, Chia-Chow K. Inhomogeneous deformation and residual stress in skin-pass axisymmetric drawing. J Mater Process Technol 2008;201:128–32.
[17] ANSYS Inc., ANSYS Release 11.0 User's Manual, 2007.
[18] Perzyna P. Fundamental problems in viscoplasticity. Advances in applied mechanics. Academic Press; 1966.
[19] Obi AI, Oyinlola AK. Frictional characteristics of fatty-based oils in wire drawing. Wear 1996;194:30–7.
[20] Dubois A, Lazzarotto L, Dubar L, Oudin J. A multi-stap lubricant evaluation strategy for wire drawing-extrusion-cold heading sequence. Wear 2002;249: 951–61.
[21] Dubar L, Dubois A, Dubar M. Friction and wear phenomena in cold metal forming: an integrated approach. In: Proceedings of international conference on tribology in manufacturing processes (ICTMP); 2004.
[22] Danishevs'kyy V, Guillet A, Pareige P, Weichert D. An analytical approach to the plastic flow of a bimetallic mono-filamentary wire through a conical die. Mech Mater 2008;40:284–92.
[23] Oeverstam H. The influence of bearing geometry on the residual stress state in cold drawn wire, analysed by the FEM. J Mater Process Technol 2006;171: 446–50.